Geometric Organization of Western and Confucian Philosophical Texts in Semantic space

Jiahong Guo
Wuhan Foreign Languages School
Wuhan, China
Jh20090128@outlook.com

***Abstract*—This study investigates whether Confucian and Western philosophical traditions exhibit different intrinsic geometries in semantic space. We propose a computational framework that encodes representative texts into high-dimensional embeddings using Sentence-BERT, followed by truncated ZCA whitening to mitigate anisotropy. Three candidate geometries—Euclidean, Spherical, and Hyperbolic—were evaluated at local and global scales. Local geometric properties were assessed via Ollivier-Ricci curvature on k-nearest neighbor (kNN) graphs, while global organization was evaluated using held-out normalized distance distortion under five-fold cross-validation. Results indicate that both Confucian and Western philosophies exhibit spherical geometry locally and hyperbolic geometry globally. Robustness tests across dimensionality, neighborhood size, and ablation study confirm these findings. This suggests that despite cultural differences, philosophical traditions share a common multiscale organizational structure characterized by local conceptual clustering and global hierarchical expansion.**

Keywords—Semantic Space; Curvature Estimation; Cross-Cultural Philosophy Comparison; Natural Language Processing; non-Euclidean geometry

I. INTRODUCTION

Modern embedding models represent semantic relationships as points in high-dimensional spaces, typically relying on Euclidean distance or cosine similarity[7]. However, while distance-based embeddings effectively capture lexical similarity, they may not fully represent philosophical concepts, which can be more structurally complex. For instance, Confucian concepts such as Dao, Qi, and De form interconnected systems, as do Western ontological concepts like Logos, Being, and Substance.

Recent developments in geometric machine learning suggest that semantic spaces need not be restricted to Euclidean geometry[8]. They can be modeled as manifolds with non-zero curvature. In this study, using a data-driven framework, we compare Euclidean, Spherical, and Hyperbolic models to evaluate which best preserves semantic relationships at both local and global scales across Confucian and Western corpora.

Primary contributions:1. Reach the conclusion that Western philosophy and Confucian philosophy both show spherical geometry locally and hyperbolic geometry globally. 2. Explore how mathematical structures can provide new insights into the organization of philosophical knowledge across cultures. 3. Find that philosophical traditions across different cultures share the same organizational structure in arrangement of ideas and content. This finding overturns our prior understanding of philosophy: It does not evolve linearly, but expands in a tree-like, hierarchal structure.

II. BACKGROUND

A. Semantic Space

Semantic space is a theoretical construct used in linguistic and natural language processing to represent the meaning and relationship of words and phrases as points or vectors within a multi-dimensional coordinate system. By mapping words and phrases using Sentence transformers to continuous vectors. Cosine similarity[1](the smaller the angle between two vectors, the closer their meanings are) between vectors approximates contextual likeness, enabling rudimentary tasks such as synonym detection or topic clustering[9]. However, traditional natural language processing (NLP), from Latent Semantic Analysis to early Word2Vec, implicitly assume a flat Euclidean geometry[10,11]. Although it can reveal some simple words relations, it might not be enough to deal with philosophy which has more complex structure.

B. Geometry (Spherical, Hyperbolic and Euclidean)

Three fundamental geometry with distinct curvatures are examined: Euclidean geometry (zero curvature), Hyperbolic geometry (constant negative curvature) and, Spherical geometry (constant positive curvature).

C. Definition of Anisotropy & How to Fix Anisotropy via ZCA Whitening in This Research.

In practical NLP applications, high-dimensional embeddings generated by pre-trained language models often exhibit anisotropy, which causes that embedding vectors are not uniformly distributed in the high-dimensional space but are heavily confined within a narrow conical region. Mathematically, this "cone effect" [12] leads to an eigenvalue spectrum that is heavily top-heavy (meaning that the vast majority of variance is concentrated in only a few dimensions). Such anisotropy severely distorts distance metrics and introduces bias into the subsequent curvature estimation of semantic manifolds.

To address this issue, this study adopts ZCA (Zero-phase Component Analysis) Whitening as a standard normalization preprocessing step. The mathematical essence of ZCA whitening is to decorrelate the data and balance the variance across dimensions while preserving the original coordinate directions through an orthogonal rotation. Its transformation matrix is defined as:

$$W_{ZCA} = U\Lambda^{-1/2}U^T \quad (1)$$

where U and Λ are the eigenvector and eigenvalue matrices of the covariance matrix. To ensure a shared reference frame, we compute a joint covariance matrix from the concatenated Confucian and Western embeddings, truncating the dimensionality to d=193 components to preserve 95% of the cumulative explained variance.

III. METHODOLOGY AND RESULTS

A. Methodology

To investigate whether Confucian and Western philosophical traditions exhibit distinct geometries in semantic space, this study implements a computational pipeline that transforms philosophical corpora into high dimensional vectors, applies global ZCA whitening to correct for anisotropy, estimates their properties at both local and global scales and subsequently aligns the spaces strictly for unified visualization. The overall workflow is illustrated in

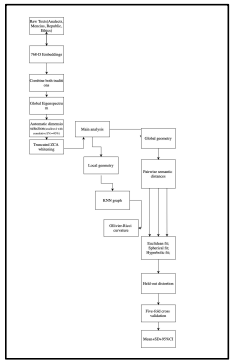


Fig. 1. Methodology Pipeline

B. Dataset Construction

10000 text excerpts were generated for both Confucian philosophy and Western philosophy. The Chinese texts are selected from 25 philosophical works including The Analects, Mencius, The Classic of Filial Piety, Laozi, Liezi and others. The Western corpus contains 6 works by Plato, four by Aristotle, plus writing from Descartes, Kant and other philosophers, amounting to 18 volumes in total.

C. Semantic Embedding

To map raw text into a computable numerical space, this study employed a pre-trained Sentence-BERT model: paraphrase-multilingual-mpnet-base-v2, which specializes in comparing texts across different languages. This model transforms each textual chunk into a 768-dimensional embedding vector.

Mathematically, for a given text chunk c, let:

$$v_c = Model(c) where\ v_c \in R^{768} \quad (2)$$

This process yields two embedding metrics: $X \in R^{10000x768}$ for the Confucian corpus and $Y \in R^{10000x768}$ for the Western corpus.

D. Automatic Dimension Selection (95% Explain Variance)

After merging the embedding matrices of Confucian and Western philosophical texts, we first conduct dimension truncation. To decide the truncated dimension, we use eigenvalues of the covariance matrix to determine the dimensions to retain. Specifically, we select the minimal dimension d that preserves at least 95% of the cumulative explained variance.

$$d = min\{k: \frac{\sum_{i=1}^{k} \lambda_i}{\sum_{i=1}^{768} \lambda_i} \geq 95\%\} \quad (3)$$

Where λ_i denotes the i-th eigenvalue of the global variance matrix and D is the original embedding dimension. This criterion preserves the dominant variance structure while removing low-variance components. The same globally derived ZCA transformation and the selected dimension d are then applied separately to the Confucian and Western embeddings, ensuring that both corpora are represented within the same geometry reference space.

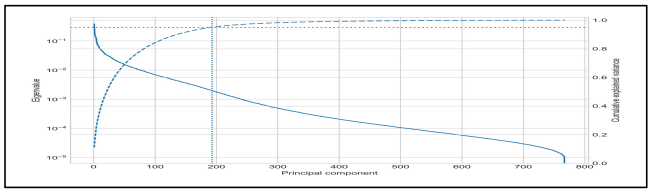


Fig. 2. Eigenvalue Spectrum (After implementing the code, the resulting value of d is 193)

E. Isotropy Transform (ZCA Whitening)

Raw sentence embeddings often exhibit a strong anisotropic cone effect , which severely distort distance-based geometric measurements, especially curvature estimation[13]. To eliminate it, global ZCA whitening transformation was applied to the concatenated embedding matrix of both traditions.

Let

$$Z = \begin{bmatrix} X \\ Y \end{bmatrix} \in R^{(10000+10000)x768} \quad (4)$$

be the combined data. Compute the mean $\mu = \frac{1}{N} x \sum_i z_i$ and covariance matrix $\sum = \frac{1}{N} \sum_i (z_i - \mu)(z_i - \mu)^T$. Then perform the eigen-decomposition $\sum = U \Lambda U^T$, where $\Lambda = diag(\lambda_1, ..., \lambda_{768})$ with $\lambda_1 \geq \lambda_2 \geq .....$

The ZCA whitening matrix is defined as:

$$W = U diag(\frac{1}{\sqrt{\epsilon + \lambda_i}}) U^T \quad (5)$$

where ϵ is a small regularization constant to avoid division by zero. In practice, this study retains only the top d=193 principal components to discard noisy high-frequency dimensions. This results in truncated ZCA whitening, which projects the whitened space down to 193 dimensions. The whitened embedding for each chunk becomes:

$$\tilde{z}_i = W_{193}(z_i - \mu), \quad (6)$$

This transformation decorrelates the features and equalizes the variance along the retained directions, effectively making the space approximately isotropic (i.e., no dominant cone direction). Consequently, subsequent geometric analyses are no longer biased by the cone structure. The transformation is fit on the combined set , and subsequently applied to the Confucian and Western embeddings individually.

F. Construction of kNN Graph for Subsequent Analysis of Local Geometric Structures.

A k-nearest neighbor graph is constructed to transform the continuous high-dimensional embedding space into a discrete graph representation, enabling the analysis of local geometric

structures. In this study, each node represents an individual philosophical text chunk, while edges represent local semantic proximity between chunks. This graph-based representation allows estimation of discrete geometric properties, including Ollivier-Ricci curvature.

The kNN graphs are constructed separately for the Confucian and Western philosophical corpora using the whitened and unaligned embeddings. Euclidean distance is adopted as the neighborhood metric. Then, This study uses a method to automatically select the k value for the k-nearest-neighbor (kNN) graph rather than fixing an arbitrary value in advance. For each philosophical tradition, up to 2,000 samples are randomly selected, and candidate values k=5,10,15.....,50 are evaluated. For each candidate k, graph connectivity is first evaluated. Let $G_{max}(k)$ denote the largest connected component of the graph and N the total number of nodes, This connectivity ratio is defined as

$$C(k)=\frac{|G_{max}(k)|}{N} \tag{7}$$

The algorithm requires $C(k)\geq 99\%$ to prevent the graph from being fragmented into isolated components. Meanwhile, the mean Ollivier-Ricci curvature $\overline{R}(k)$ is calculated for each k, and curvature stability is evaluated suing the change between consecutive k values:

$$\triangle R(k)=|\overline{R}(\mathrm{k})-\overline{R}(k-\triangle k)| \tag{8}$$

Curvature is considered to have reached a stability when $\triangle R(k)\leq 0.2$ for two consecutive steps. For each philosophical tradition, the minimum k satisfying the connectivity criterion, $k_{connectivity}$, and the minimum k satisfying the curvature-stability criterion, $k_{stability}$, are identified. The candidate value for each tradition is then chosen as $\mathrm{k}=max(k_{connectivity}, k_{stability})$. Finally, to ensure that different philosophical traditions are compared using the same neighborhood scale, the largest candidate value across all traditions is selected as the final k. After implementation of the code, we finally set k=10 as the final value for constructing the k-NN graph.

G. Local Curvature Estimation (Ollivier-Ricci curvature)

To characterize the local geometric structure of philosophical semantic spaces, this study employs a discrete curvature measure on the constructed kNN graphs: Ollivier-Ricci curvature.

Ricci curvature, originally developed in differential geometry, describes how the geometry of a space deviates from flatness by measuring the local expansion or contraction of neighboring regions. While classical Ricci curvature is defined on smooth manifolds, many real-world data structures, including semantic networks and high-dimensional embedding spaces, are naturally represented as discrete graphs. Ollivier-Ricci curvature extends the concept of Ricci curvature to graph structures by interpreting the relationship between neighboring nodes through optimal transport. In the context of semantic embedding spaces, Ollivier-Ricci curvature [5] provides a graph-based approach for quantifying local semantic organization.

Ollivier-Ricci curvature measures the degree of similarity between the local neighborhoods of two connected nodes in a graph. For an edge connecting two nodes x and y, the curvature is defined as:

$$\kappa(x,y)=1-\frac{W_1(m_x,m_y)}{d(x,y)} \tag{9}$$

where d(x,y) represents the distance between the two nodes, m_x and m_y denote the local probability distributions centered at x and y, and W_1 is the Wasserstein-1 distance between these distributions. Positive curvature indicates spherical geometry with clustering properties, whereas negative curvature suggests hyperbolic geometry with an expanding, tree-like neighborhood structure. In this study, Ollivier-Ricci curvature is used to evaluate whether local semantic neighborhoods exhibit clustering or hierarchical expansion patterns.

After running the code, the computed mean Olivier Ricci curvature is 0.1877 for Confucian philosophy and 0.1224 for Western philosophy. This suggests that, to a certain extent, both philosophical corpora exhibit spherical geometry in their local geometry.

H. Global geometry estimation (held-out distortion & 5-fold cross validation)

To evaluate the overall geometric structure of the philosophical semantic spaces, this study performs global geometry estimation by comparing three candidate geometric models: Euclidean, spherical, and hyperbolic spaces. Unlike local curvature measures that characterize neighborhood-level structures, global geometry estimation evaluates whether the entire embedding space is better represented by a specific geometry.

For each philosophical tradition, the quality of each geometric model is measured using distortion, which quantifies the discrepancy between the original embedding-space distances and the distances reconstructed under a candidate geometry. Specifically, the distortion is calculated as the normalized difference between between the original pairwise distances d_{ij} and the fitted geometric distances$\widehat{d_{ij}}$:

$$Distortion=\frac{\sum_{i,j}|d_{i,j}-\hat{d}_{i,j}|}{\sum_{i,j}d_{i,j}} \tag{10}$$

where $d_{i,j}$ denotes the observed distance between two embeddings and $\hat{d}_{i,j}$ represents the corresponding distance after fitting the candidate geometric model. Lower distortion indicates that the geometry provides a more accurate representation of the semantic space.

To evaluate whether the estimated geometry generalizes beyond the fitted samples, a held-out distortion evaluation[6] is performed. The dataset is divided into training and test subsets, where geometric parameters are estimated using the training subset and subsequently evaluated on unseen samples. This procedure reduces the possibility that a geometric model is selected merely due to overfitting.

Furthermore, five-fold cross-validation is conducted to assess the robustness and stability of the geometric estimation. The embeddings are randomly partitioned into five subsets, with each fold using four subsets for model fitting and the remaining subset for evaluation. The final geometric performance is reported as the mean distortion across five folds with corresponding variance measures. A consistently lower held-out distortion across folds indicates stronger evidence that the corresponding geometry captures the underlying organization of the semantic space.

After code implementation, we find that across all validation folds, hyperbolic geometry consistently achieved the lowest reconstruction distortion for both philosophical traditions. For Confucian philosophy, hyperbolic geometry achieved a mean cross-validated distortion of 0.7405 with a standard deviation of 0.0161 (95% CI: [0.7205, 0.7605]). Similarly, the Western semantic space showed a preference for hyperbolic geometry, with a mean distortion of 0.9048, SD = 0.0048 (95% CI: [0.8988, 0.9108]).

The relatively small standard deviations and narrow confidence intervals indicate that the preference for hyperbolic geometry is stable across different data partitions rather than being driven by a specific train-test split. These results suggest that hyperbolic geometry provides the most effective global representation of the semantic organization in both philosophical traditions.

I. Manifold Alignment and Visualization

Cross-cultural semantic visualization faces a challenge: independently trained embeddings may have arbitrary coordinate rotations or reflections despite similar underlying structures. Direct visualization using native coordinates can obscure structural comparisons.

To enable visual comparison, the Procrustes Analysis[2] was applied as a post-processing alignment step. The procedure estimates an optimal transformation that maps one embedding space onto the other by minimizing the reconstruction error between the two point clouds.

Following alignment, Uniform Manifold Approximation and Projection (UMAP)[3] was applied to obtain two-dimensional representations for visualization. UMAP was selected because it preserves local neighborhood relationships while maintaining aspects of the global structure[4] of high-dimensional data, making it suitable for visualizing complex semantic manifolds.

Importantly, the alignment and dimensionality reduction procedures serve only for visualization. All geometric analyses are performed independently.

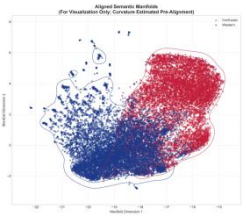


Fig. 3.Visualization

J. Robustness Analysis

1) Dimensionality Sensitivity Test

a) Methodology: The dimensionality reduction step following ZCA whitening introduces a potential methodological choice, as retaining different numbers of dimensions may influence the geometric structure. Although the main analysis retains dimensions explaining 95% of the total variance (193 dimensions), a sensitivity analysis was conducted to evaluate whether the observed geometric patterns depend on this specific truncation threshold.

Specifically, the ZCA-whitened embeddings were reconstructed under alternative dimensionality settings, and the subsequent geometric analyses were repeated using these representations. The robustness of the results was evaluated by comparing whether the preferred global geometry and local geometric measurements remained consistent across different retained dimensions.

b) Result: The dimensionality sensitivity test demonstrated that the local geometric measurements remained highly stable across different retained dimensions.

The data shows that the estimated Ollivier-Ricci curvature values remained relatively consistent across all dimensional settings. For the Confucian philosophy, the mean Ricci curvature values: C_{Ricci}=0.1712-0.2087. Similarly, the Western philosophy showed stable estimates: W_{Ricci}=0.1114-0.1522. Specifically, at the primary setting of 193 dimensions (approximately 95% explained variance)**,** the curvature estimates were: C_{Ricci}=0.1877 and W_{Ricci}=0.1224 which are highly consistent with results obtained under alternative dimensional configurations. These results indicate that the observed local geometric properties are not caused by the specific choice of the 193-dimensional truncation.

2) kNN Neighborhood Size Robustness Test

a) Methodology: In the main analysis, k=10 was selected. To evaluate whether the curvature estimation was sensitive to this parameter choice, additional experiments were conducted using k=5,10, 20 and 30. For each value of k, the kNN graphs were reconstructed from the same ZCA-whitened embeddings, and Ollivier-Ricci curvature was recalculated.

b) Result: The kNN robustness analysis (Table 1) demonstrated that the observed positive curvature structure remained stable across different neighborhood scales. When varying k from 5 to 30, the mean Ricci curvature values changed moderately but consistently remained positive.

Table I: K Robustness Test

K	C_Ricci	W_Ricci
5	+0.1978	+0.1627

10	+0.1877	+0.1224
20	+0.1879	+0.1143
30	+0.1952	+0.1161

3) Ablation Study on ZCA Whitening

a) Methodology: To examine whether the observed spherical geometry is an intrinsic property of the representation space or an artifact introduced by preprocessing, we conducted an ablation study by removing ZCA whitening from the pipeline. All other experimental settings remained identical to those in the main experiment.

In the original analysis, ZCA whitening was applied. However, whitening itself transforms the covariance structure of representation space and could potentially influence the local curvature measurement. Therefore, comparing the results with and without whitening provides a test of the robustness of the observed geometric pattern.

If the spherical pattern remains after removing ZCA whitening, this would indicate that the observed positive local curvature is not solely attributable to the whitening transformation and is therefore more consistent with an underlying property of the original representation space.

b) Result: The Ricci curvature for Confucian philosophy is 0.0879 and for Western philosophy is 0.0978, which indicates that without ZCA whitening, their intrinsic geometry remains spherical. This results proves that the observed geometry is not sensitive to the preprocessing procedure and is not an artifact.

4) Validation of Global Geometry Result.

a) Methodology: To examine whether the hyperbolic geometry observed in philosophical texts is specific to philosophical discourse rather than a general property of text representations, we introduce non-philosophical texts as control groups. Since hyperbolic spaces are well suited to representing hierarchical structures, the hyperbolic pattern observed in philosophical texts could potentially arise from the organized and hierarchical nature of the text itself.

We compare the global geometric properties of philosophical and non-philosophical texts while keeping other things identical. If some non-philosophical texts do not exhibit hyperbolic structure, this would suggest that the observed hyperbolicity is not a universal property of textual representations.

b) Result: When we replace philosophical texts with news texts, we see one category is Euclidean. This shows the hyperbolicity we observed is not universal for structured text representations. And we can also deduce that within a shared representation space, different corpora can exhibit distinct global geometric organizations. Since the same embedding model produces different geometric patterns across corpora, we can exclude the probability of measuring the geometry imposed by the embedding model.

Hyperbolic (mean=0.9726, SD=0.0129, 95% CI=[0.9566, 0.9886])

Euclidean (mean =0.9616, SD=0.0203, 95% CI=[0.9367, 0.9871])

IV. Discussion

A. Limitation

Several methodological factors may introduce potential biases that warrant consideration: First, the kNN graph construction relies on Euclidean distance, which may introduce a slight bias toward flat geometry; however, the detection of spherical local curvature despite this bias strengthens the result. Second, ZCA whitening assumes the embedding space can be normalized via Euclidean statistics.

B. The Practical Implications of this Study for Philosophy

On the local geometric level, both Western and Confucian philosophy show spherical geometry. At the global geometric level, hyperbolic geometry yields the best fit for both. These findings suggest that philosophical thought may possess a similar organization pattern. At a local level, philosophical ideas tend to gather around closely related questions. Yet at a broader level, these communities do not remain isolated; they branch into different questions, concepts, and traditions while still remaining connected within a larger intellectual landscape. This suggests that philosophy is not organized as a collection of completely separate ideas, nor as a single linear system. Instead, it may organize in a system in which closely related ideas form local intellectual clusters, while these clusters gradually diverge and connect to form a broader and more complex structure of thought. The emergence of this pattern in both Western and Eastern philosophical corpora suggests that, despite substantial differences in their concepts and historical contexts, philosophical traditions may share a common way of organizing ideas.

References

[1] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," *arXiv preprint arXiv:1908.10084*, 2019, doi: 10.48550/arXiv.1908.10084.

[2] R. Hurley and R. B. Cattell, "The Procrustes program: Producing direct rotation to test a hypothesized factor structure," *Behavioral Science*, vol. 7, no. 2, pp. 258–262, 1962.

[3] L. McInnes, J. Healy, and J. Melville, "UMAP: Uniform manifold approximation and projection for dimension reduction," *arXiv preprint arXiv:1802.03426*, 2018.

[4] H. Huang, Y. Wang, C. Rudin, and E. P. Browne, "Towards a comprehensive evaluation of dimension reduction methods for transcriptomic data visualization," *Communications Biology*, vol. 5, no. 1, Art. no. 719, 2022.

[5] M. Eidi and J. Jost, "Ollivier Ricci curvature of directed hypergraphs," *arXiv preprint arXiv:1907.04727*, 2019, doi: 10.48550/arXiv.1907.04727.

[6] T. Abedin, H. Xu, and S. Uddin, "The impact of K selection in K-fold cross-validation on bias and variance in supervised learning models," *Scientific Reports*, vol. 16, Art. no. 6084, 2026.

[7] T. B. Hashimoto, D. Alvarez-Melis, and T. S. Jaakkola, "Word embeddings as metric recovery in semantic spaces," *Transactions of the Association for Computational Linguistics*, vol. 4, pp. 273–286, 2016, doi: 10.1162/tacl_a_00098.

[8] M. Nickel and D. Kiela, "Poincaré embeddings for learning hierarchical representations," *arXiv preprint arXiv:1705.08039*, 2017, doi: 10.48550/arXiv.1705.08039.

[9] K. You, "Semantics at an angle: When cosine similarity works until it doesn't," *arXiv preprint arXiv:2504.16318*, 2025, doi: 10.48550/arXiv.2504.16318.

[10] M. N. Jones, T. M. Gruenenfelder, and G. Recchia, "In defense of spatial models of semantic representation," *New Ideas in Psychology*, vol. 50, pp. 54–60, 2018, doi: 10.1016/j.newideapsych.2017.08.001.

[11] T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and G. S. Dean, "Distributed representations of words and phrases and their compositionality," in *Proc. 26th Int. Conf. Neural Information Processing Systems (NIPS)*, 2013, pp. 3111-3119.

[12] T. Wollschläger, J. Elstner, S. Geisler, V. Cohen-Addad, S. Günnemann, and J. Gasteiger, "The geometry of refusal in large language models: Concept cones and representational independence," in *Proc. 42nd Int. Conf. Machine Learning (ICML)*, vol. 267, 2025, pp. 66945–66970.

[13] R. Bernas, F. Jourdan, A. Poché, and C. Hudelot, "Revisiting anisotropy in language transformers: The geometry of learning dynamics," *arXiv preprint arXiv:2604.08764*, 2026, doi: 10.48550/arXiv.2604.08764.